

Presentation of top VTs

Erland Brandser Olsson

2025

Time series

- ▶ Historical raw material deliveries can be used for time series forecasting.
- ▶ Raw material deliveries collected from irregular time intervals.
- ▶ One solution is to resample irregular timestamps to fixed intervals.

Idea

- ▶ The material deliveries might follow time series patterns.
- ▶ One series of raw material deliveries over time can alone encode meaningful information; ordering-schedule, ordering-volume, ordering-variability, ordering-movement, high and low seasons etc.

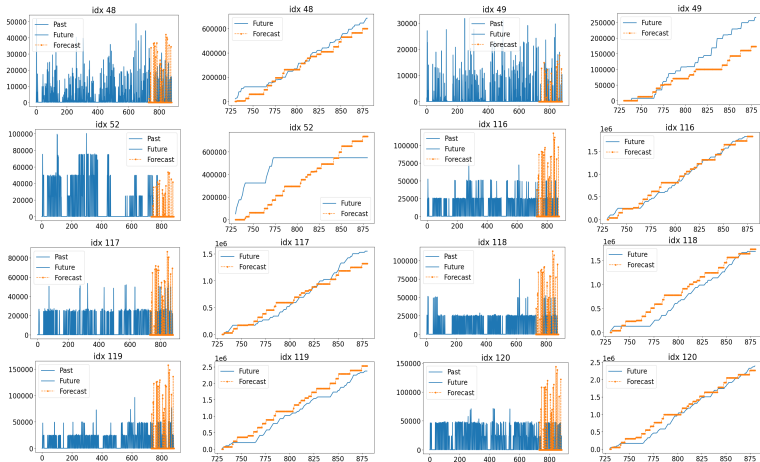
Neural forecasting

- ▶ Neural forecasting methods have in recent years accelerated and now shows strong performance in competitions and industrial systems [1].
- ▶ These methods can, to a much bigger extent than classical ML methods, automate the feature-engineering process and learn the right features from raw data.
- ▶ Learning features can give better generalization and can save a lot of time and effort if you frequently are given very different forecasting-problems.
- ▶ This can also avoid building ad-hoc solutions with hand-crafted features, that don't have guarantees to work on other forecasting problems., e.g. for each year the VTs might have to use completely different approaches and features.

Neural forecasting regressor

- ▶ NHiTS [1] is a state-of-the-art neural forecasting method which work well on difficult univariate time series forecasting problems over many timesteps. NHiTS builds on NBeats [3] which was referenced in the time-series lecture notes.
- ▶ VT1 is a NHiTS network. Trained with a validation set from 2024-07-22 to 2024-12-19 used for early-stopping, checkpoint-callbacks and reducing learning rate on plateau.
- ▶ Ensemble of different architectures give better robustness; VT1 combines the NHiTS forecast with the forecast from one recurrent GRU [2] neural network. It is possible to improve this even further with more complex ensemble techniques.

Raw material deliveries and aggregated sum on validation set from 2024-07-22 to 2024-12-19



- [1] Cristian Challu et al. “Nhits: Neural hierarchical interpolation for time series forecasting”. In: *Proceedings of the AAAI conference on artificial intelligence*. Vol. 37. 6. 2023, pp. 6989–6997.
- [2] Kyunghyun Cho et al. “On the Properties of Neural Machine Translation: Encoder–Decoder Approaches”. In: *Proceedings of SSST-8, Eighth Workshop on Syntax, Semantics and Structure in Statistical Translation*. Ed. by Dekai Wu et al. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 103–111. DOI: 10.3115/v1/W14-4012. URL: <https://aclanthology.org/W14-4012/>.
- [3] Boris N. Oreshkin et al. “N-BEATS: Neural basis expansion analysis for interpretable time series forecasting”. In: *International Conference on Learning Representations*. 2020. URL: <https://openreview.net/forum?id=r1ecqn4YwB>.